

THE MASTER DIFFERENTIAL REFERENCE · ESSENTIAL CALCULUS FOR DIFFERENTIAL EQUATIONS

Calculus for ODEs: Derivatives & The Differential Engine

Operational differentiation rules, transcendental function catalog, implicit & multivariable forms, and their direct structural mechanics in solving Ordinary Differential Equations.

No Formula Sheet Permitted on Exams

Exam Speed & Accuracy

Reverse Product Rule = Integrating Factor

Zill Chapters 1, 2, 3, 5

1. CORE OPERATIONAL DIFFERENTIATION RULES

Algebraic Foundation

POWER & LINEARITY RULES

BASIC

Power: $\frac{d}{dx}[x^n] = nx^{n-1} \quad (n \in \mathbb{R})$

Constant: $\frac{d}{dx}[c] = 0, \quad \frac{d}{dx}[cf] = cf'$

Sum / Diff: $\frac{d}{dx}[f \pm g] = f'(x) \pm g'(x)$

PRODUCT & QUOTIENT RULES

HIGH-YIELD ODE

Product: $\frac{d}{dx}[u \cdot v] = u'v + uv'$

Quotient: $\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$

Reciprocal: $\frac{d}{dx}\left[\frac{1}{v}\right] = -\frac{v'}{v^2}$

CHAIN RULE & DIFFERENTIALS

SUBSTITUTIONS

Chain: $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$

Leibniz: $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$

Differential: $dy = y'(x) dx \iff \frac{dy}{dx} = y'$

2. THE MASTER TRANSCENDENTAL FUNCTIONS CATALOG

Memorize Exactly

EXPONENTIAL & LOGARITHMIC

NATURAL & GENERAL

$\frac{d}{dx}[e^u] = e^u \cdot u'$

$\frac{d}{dx}[a^u] = a^u \ln(a) \cdot u' \quad (a > 0)$

$\frac{d}{dx}[\ln|u|] = \frac{u'}{u} \quad (u \neq 0)$ **[Top ODE Tool]**

$\frac{d}{dx}[\log_a|u|] = \frac{u'}{u \ln(a)}$

$\frac{d}{dx}[u^v] = u^v \left(v' \ln u + \frac{vu'}{u} \right)$

TRIGONOMETRIC FUNCTIONS

OSCILLATIONS & MODELS

$\frac{d}{dx}[\sin u] = \cos(u) \cdot u'$

$\frac{d}{dx}[\cos u] = -\sin(u) \cdot u'$

$\frac{d}{dx}[\tan u] = \sec^2(u) \cdot u'$

$\frac{d}{dx}[\cot u] = -\csc^2(u) \cdot u'$

$\frac{d}{dx}[\sec u] = \sec(u) \tan(u) \cdot u'$

$\frac{d}{dx}[\csc u] = -\csc(u) \cot(u) \cdot u'$

INVERSE TRIG & HYPERBOLIC

ROOTS & CATENARIES

$\frac{d}{dx}[\arcsin u] = \frac{u'}{\sqrt{1-u^2}} \quad (|u| < 1)$

$\frac{d}{dx}[\arccos u] = -\frac{u'}{\sqrt{1-u^2}}$

$\frac{d}{dx}[\arctan u] = \frac{u'}{1+u^2}$ **[Separable staple]**

$\frac{d}{dx}[\sinh u] = \cosh(u) \cdot u' \quad (\sinh x = \frac{e^x - e^{-x}}{2})$

$\frac{d}{dx}[\cosh u] = \sinh(u) \cdot u' \quad (\cosh x = \frac{e^x + e^{-x}}{2})$

$\frac{d}{dx}[\tanh u] = \operatorname{sech}^2(u) \cdot u', \quad \cosh^2 u - \sinh^2 u = 1$

3. MULTIVARIABLE, IMPLICIT & HIGHER-ORDER DERIVATIVES

Exactness & Verification

IMPLICIT DIFFERENTIATION (SECTION 1.1)

VERIFICATION

To find $\frac{dy}{dx}$ from an implicit relation $G(x, y) = 0$:

$$\frac{d}{dx}[G(x, y)] = \frac{\partial G}{\partial x} + \frac{\partial G}{\partial y} \frac{dy}{dx} = 0$$
$$\implies \frac{dy}{dx} = -\frac{G_x(x, y)}{G_y(x, y)} \quad (G_y \neq 0)$$

Crucial for verifying implicit general solutions $F(x, y) = C$ without solving for $y(x)$.

TOTAL DIFFERENTIAL & PARTIALS (SECTION 2.4)

EXACT ODES

The total differential of a surface $f(x, y)$:

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy$$

If $df = 0$, then $f(x, y) = C$ (level curves).

Clairaut's Schwarz Theorem:

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} \iff \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

HIGHER DERIVATIVES & LEIBNIZ RULE

SERIES & ORDER

Product rule for second & third derivatives:

$$(uv)'' = u''v + 2u'v' + uv''$$
$$(uv)''' = u'''v + 3u''v' + 3u'v'' + uv'''$$

General Leibniz Formula:

$$(uv)^{(n)} = \sum_{k=0}^n \binom{n}{k} u^{(n-k)} v^{(k)}$$

Powers Reduction of Order $y_2 = uy_1$ and Cauchy-Euler equations.

⚡ SYSTEM 1: THE ODE DERIVATIVE BRIDGE — HOW DERIVATIVES POWER ODE SOLUTIONS

DIRECT MIDTERM 1 & 2 APPLICATION

1. Integrating Factor Reverse Product Rule

Zill Section 2.3

ODE Standard Form: $y' + P(x)y = Q(x)$. Multiply by $\mu(x) = e^{\int P(x)dx}$:

$$\mu(x)y' + P(x)\mu(x)y = \mu(x)Q(x)$$

The Derivative Miracle: Since $\mu'(x) = P(x)\mu(x)$, the LHS collapses exactly:

$$\frac{d}{dx}[\mu(x) \cdot y] = \mu(x)Q(x) \iff \text{Product Rule!}$$

3. Bernoulli Substitution via Chain Rule

Zill Section 2.5

ODE: $y' + P(x)y = Q(x)y^n \implies y^{-n}y' + P(x)y^{1-n} = Q(x)$.

Chain Rule Transformation: Let $u = y^{1-n}$. Then:

$$\frac{du}{dx} = (1-n)y^{-n} \frac{dy}{dx} \implies y^{-n}y' = \frac{1}{1-n} \frac{du}{dx}$$

Yields the linear ODE: $\frac{du}{dx} + (1-n)P(x)u = (1-n)Q(x)$.

5. Characteristic Eq via Exponential Derivatives

Zill Section 3.3

ODE: $ay'' + by' + cy = 0$. Test exponential trial $y = e^{rx}$.

Derivatives: $y' = re^{rx}$ and $y'' = r^2e^{rx}$. Factoring out $e^{rx} \neq 0$:

$$(ar^2 + br + c)e^{rx} = 0 \iff ar^2 + br + c = 0 \quad (\text{Algebraic Poly!})$$

2. Exactness Condition via Mixed Partials

Zill Section 2.4

ODE Differential Form: $M(x, y)dx + N(x, y)dy = 0$.

Derivative Test: If there exists a potential function $f(x, y)$ such that $df = Mdx + Ndy$, then $f_x = M$ and $f_y = N$.

$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial N}{\partial x} \implies \text{Exact iff } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

4. Homogeneous Substitution via Product Rule

Zill Section 2.5

ODE: $\frac{dy}{dx} = g(y/x)$. Let $y = u \cdot x$ where $u = u(x)$.

Product Rule Differentiation:

$$\frac{dy}{dx} = \frac{d}{dx}[u \cdot x] = u \cdot \frac{dx}{dx} + x \frac{du}{dx} = u + x \frac{du}{dx}$$

Substitute to separate variables: $x \frac{du}{dx} = g(u) - u \implies \frac{du}{g(u)-u} = \frac{dx}{x}$.

6. Reduction of Order via Leibniz Product Rule

Zill Section 3.7

Known solution: $y_1(x)$ to $y'' + P'y' + Qy = 0$. Set $y_2(x) = u(x)y_1(x)$.

Product Derivatives: $y_2' = u'y_1 + uy_1'$ and $y_2'' = u''y_1 + 2u'y_1' + uy_1''$.

When plugged in, all u terms sum to zero: $y_1u'' + (2y_1' + Py_1)u' = 0$ (1st order in $w = u'$).

THE MASTER INTEGRAL REFERENCE · ESSENTIAL CALCULUS FOR DIFFERENTIAL EQUATIONS

Calculus for ODEs: Integrals & The Integration Engine

Fundamental antiderivative table, the 4 pillar integration techniques for differential equations, and their systematic deployment in solving Separable, Linear, Exact, and Higher-Order ODEs.

Critical ODE Skill: Rapid Antiderivative Recognition Tabular IBP & Partial Fractions Always Include +C Before Dividing! Zill Chapters 1, 2, 3

1. MASTER ANTIDERIVATIVE CATALOG

Standard Forms

POWER & EXPONENTIALS

CORE

$\int u^n du = \frac{u^{n+1}}{n+1} + C \quad (n \neq -1)$

$\int \frac{1}{u} du = \ln|u| + C \quad (u \neq 0)$

$\int \frac{g'(x)}{g(x)} dx = \ln|g(x)| + C$ [Top Shortcut]

$\int e^{ku} du = \frac{1}{k} e^{ku} + C \quad (k \neq 0)$

$\int a^u du = \frac{a^u}{\ln a} + C \quad (a > 0, a \neq 1)$

$\int \ln(u) du = u \ln(u) - u + C$

TRIGONOMETRIC INTEGRALS

TRIG STANDARDS

$\int \cos(u) du = \sin(u) + C$

$\int \sin(u) du = -\cos(u) + C$

$\int \sec^2(u) du = \tan(u) + C$

$\int \csc^2(u) du = -\cot(u) + C$

$\int \sec(u) \tan(u) du = \sec(u) + C$

$\int \tan(u) du = \ln|\sec(u)| + C = -\ln|\cos(u)| + C$

$\int \cot(u) du = \ln|\sin(u)| + C$

$\int \sec(u) du = \ln|\sec(u) + \tan(u)| + C$

QUADRATIC & INVERSE TRIG

QUADRATIC DENOMINATORS

$\int \frac{du}{u^2+a^2} = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C$

$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin\left(\frac{u}{a}\right) + C$

$\int \frac{du}{u^2-a^2} = \frac{1}{2a} \ln\left|\frac{u-a}{u+a}\right| + C \quad (u^2 > a^2)$

$\int \frac{du}{a^2-u^2} = \frac{1}{2a} \ln\left|\frac{a+u}{a-u}\right| + C \quad (u^2 < a^2)$

$\int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln|u + \sqrt{u^2 \pm a^2}| + C$

$\int \frac{u du}{u^2+a^2} = \frac{1}{2} \ln(u^2+a^2) + C$

2. THE 4 PILLAR INTEGRATION TECHNIQUES FOR DIFFERENTIAL EQUATIONS

Master Algorithms

PILLAR 1: U-SUBSTITUTION & THE LOGARITHMIC SHORTCUT

ESSENTIAL

The Golden ODE Shortcut: If the numerator is proportional to the denominator's derivative, integrate directly to a logarithm:

$\int \frac{P(x)}{Q(x)} dx = \frac{k}{m} \ln|Q(x)| + C \quad \text{when } P(x) = \frac{k}{m} Q'(x)$

Exam Examples:

- $\int \frac{x}{x^2+4} dx = \frac{1}{2} \ln(x^2+4) + C$ (Integrating Factor for $y' + \frac{x}{x^2+4}y = \dots$)
- $\int \frac{\cos x}{2+\sin x} dx = \ln(2+\sin x) + C$ (Separable ODE $y' = (2+\sin x)y$)

PILLAR 2: INTEGRATION BY PARTS & THE TABULAR (DI) METHOD

HIGH SPEED

Standard Formula: $\int u dv = uv - \int v du$ (LIATE Priority: Log, InvTrig, Alg, Trig, Exp).

The Tabular (DI) Method: For $\int (\text{poly}) \times e^{ax} dx$ or $\int (\text{poly}) \times \sin(bx) dx$:

D (Derivatives): $x^2 \rightarrow 2x \rightarrow 2 \rightarrow 0$

I (Integrals): $e^{3x} \rightarrow \frac{1}{3}e^{3x} \rightarrow \frac{1}{9}e^{3x} \rightarrow \frac{1}{27}e^{3x}$

Result across diagonals:
 $\int x^2 e^{3x} dx = \frac{x^2}{3}e^{3x} - \frac{2x}{9}e^{3x} + \frac{2}{27}e^{3x} + C$
Eliminates 3 rounds of tedious IBP algebra!

PILLAR 3: PARTIAL FRACTION DECOMPOSITION (PFD) SEPARABLE & LOGISTIC

Distinct Linear: $\frac{1}{(y-a)(y-b)} = \frac{A}{y-a} + \frac{B}{y-b}$. Heaviside Cover-up: $A = \frac{1}{a-b}, B = \frac{1}{b-a}$.

Repeated Linear: $\frac{1}{(y-a)^2(y-b)} = \frac{A}{y-a} + \frac{B}{(y-a)^2} + \frac{C}{y-b}$.

Irreducible Quadratic: $\frac{1}{(y-a)(y^2+k^2)} = \frac{A}{y-a} + \frac{By+C}{y^2+k^2}$.

The Logistic ODE Staple (Section 2.8): $\frac{dy}{y(M-y)} = kdt \implies \frac{1}{M} \int \left(\frac{1}{y} + \frac{1}{M-y} \right) dy = \int kdt$
 $\implies \frac{1}{M} [\ln|y| - \ln|M-y|] = kt + C \implies \frac{y}{M-y} = C_1 e^{Mkt}$

PILLAR 4: COMPLETING THE SQUARE & HALF-ANGLES

TRANSFORMS & FORCING

Completing the Square: $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)$.

$\int \frac{dx}{x^2+6x+25} = \int \frac{dx}{(x+3)^2+16} = \frac{1}{4} \arctan\left(\frac{x+3}{4}\right) + C$

Half-Angle Identities for Periodic Forcing Terms:

$\cos^2(x) = \frac{1+\cos(2x)}{2}, \quad \sin^2(x) = \frac{1-\cos(2x)}{2}$

$\sin(A)\cos(B) = \frac{1}{2}[\sin(A-B) + \sin(A+B)]$

SYSTEM 2: THE ODE INTEGRAL BRIDGE — HOW INTEGRALS POWER ODE SOLUTIONS

EXAM EXECUTION PROTOCOL

1. Separable Equations: Dual Integration

Zill Section 2.2

ODE: $\frac{dy}{dx} = g(x)h(y) \implies \frac{1}{h(y)} dy = g(x) dx$.

Integration: $\int \frac{1}{h(y)} dy = \int g(x) dx + C$.

Trap: Always combine arbitrary constants into a single +C on the independent variable (x) side immediately before algebraic inversion.

3. Exact Equation Potential Reconstruction

Zill Section 2.4

Step 1: Integrate $M(x, y)$ with respect to x (y is held constant):

$f(x, y) = \int M(x, y) dx + g(y)$

Step 2: Differentiate w.r.t y to find $g'(y)$: $\frac{\partial f}{\partial y} = \frac{\partial}{\partial y} \int M dx + g'(y) = N(x, y)$.

Step 3: Integrate $g'(y)$ w.r.t y : $g(y) = \int g'(y) dy$. Solution: $f(x, y) = C$.

5. Variation of Parameters Integrals

Zill Section 3.5

ODE: $y'' + P(x)y' + Q(x)y = g(x)$. Complementary: $y_c = c_1 y_1 + c_2 y_2$.

Wronskian: $W(x) = y_1 y_2' - y_1' y_2 \neq 0$.

$u_1(x) = -\int \frac{y_2 g}{W} dx, \quad u_2(x) = \int \frac{y_1 g}{W} dx$

Particular Solution: $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$.

2. Integrating Factor: Two-Stage Integration

Zill Section 2.3

Stage 1 (Integrating Factor): $\mu(x) = \exp\left(\int P(x) dx\right)$. Set $+C = 0$.

Stage 2 (General Solution): Integrate the product rule contraction:

$y(x) = \frac{1}{\mu(x)} \left[\int \mu(x) Q(x) dx + C \right]$

Fatal Mistake: Never write $y = \frac{\int \mu Q dx}{\mu} + C$. The +C MUST be inside brackets!

4. Picard's Successive Approximations

Zill Section 1.2

IVP: $y' = f(x, y), \quad y(x_0) = y_0$.

Integral Recurrence Equation:

$y_{n+1}(x) = y_0 + \int_{x_0}^x f(t, y_n(t)) dt$

Each integral generates the next order Taylor polynomial of the true solution!

6. Fundamental Theorem & Leibniz Rule

Theory & IVPs

Derivative of a Definite Integral (FTC):

$\frac{d}{dx} \left[\int_a^x f(t) dt \right] = f(x)$

General Leibniz Rule with Variable Limits:

$\frac{d}{dx} \left[\int_{u(x)}^{v(x)} f(t) dt \right] = f(v(x))v'(x) - f(u(x))u'(x)$